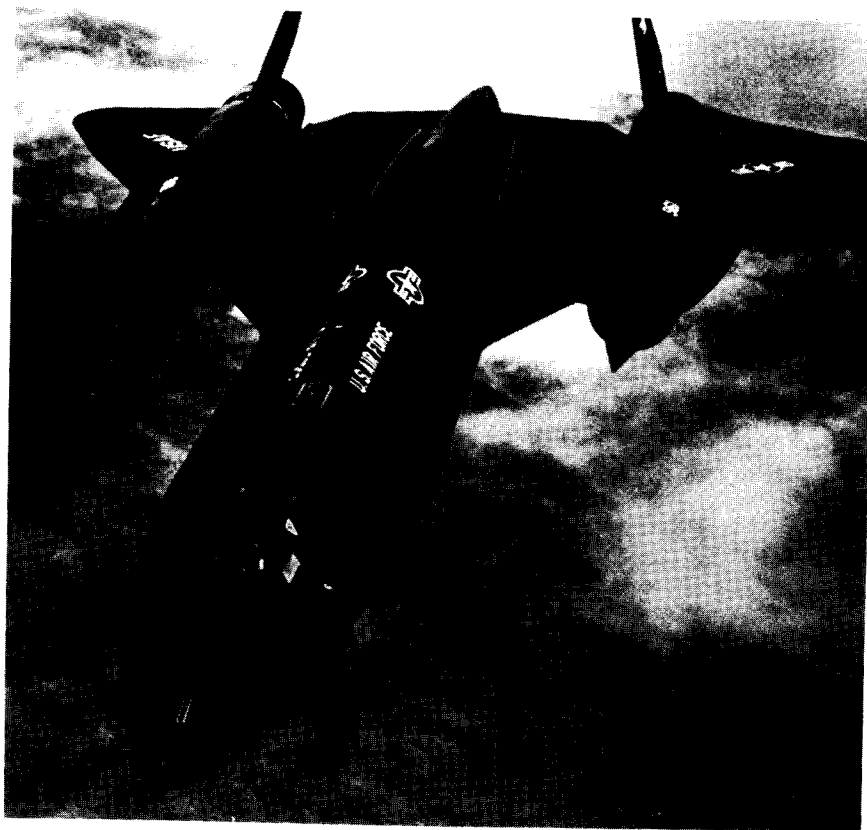


aircraft centerline. This permits connecting the left- and right-hand elevator surfaces with a torque tube, which reduces elevator flutter tendencies.

Some aircraft have no separate elevator. Instead, the entire horizontal tail is mounted on a spindle to provide variable tail incidence. This provides outstanding "elevator" effectiveness but is somewhat heavy. Some general-aviation aircraft use such an all-moving tail, but it is most common for supersonic aircraft, where it can be used to trim the rearward shift in aerodynamic center that occurs at supersonic speeds.

A few aircraft such as the SR-71 have used all-moving vertical tails to increase control authority.



Lockheed SR-71

CONFIGURATION LAYOUT AND LOFT

7.1 INTRODUCTION

The process of aircraft conceptual design includes numerous statistical estimations, analytical predictions, and numerical optimizations. However, the product of aircraft design is a drawing. While the analytical tasks are vitally important, the designer must remember that these tasks serve only to influence the drawing, for it is the drawing alone that ultimately will be used to fabricate the aircraft.

All of the analysis efforts to date were performed to guide the designer in the layout of the initial drawing. Once that is completed, a detailed analysis can be conducted to resize the aircraft and determine its actual performance. This is discussed in Chapters 12-19.

This detailed analysis is time-consuming and costly, so it is essential that the initial drawing be credible. Otherwise, substantial effort will be wasted upon analyzing an unrealistic aircraft.

This chapter and Chapters 8-11 discuss the key concepts required to develop a credible initial drawing of a conceptual aircraft design. These concepts include the development of a smooth, producible, and aerodynamically acceptable external geometry, the installation of the internal features such as the crew station, payload, landing gear, and fuel system, and the integration of the propulsion system.

Real-world considerations which must be met by the design include the correct relationship between the aerodynamic center and the center of gravity, the proper amount of pilot outside visibility, and sufficient internal access for production and maintenance.

7.2 END PRODUCTS OF CONFIGURATION LAYOUT

The outputs of the configuration layout task will be design drawings of several types as well as the geometric information required for further analysis.

The design layout process generally begins with a number of conceptual sketches. Figure 7.1 illustrates an actual, unretouched sketch from a fighter conceptual design study (Ref. 13). As can be seen, these sketches are crude and quickly done, but depict the major ideas which the designer intends to incorporate into the actual design layout.

A good sketch will show the overall aerodynamic concept and indicate the locations of the major internal components. These should include the landing gear, crew station, payload or passenger compartment, propulsion sys-

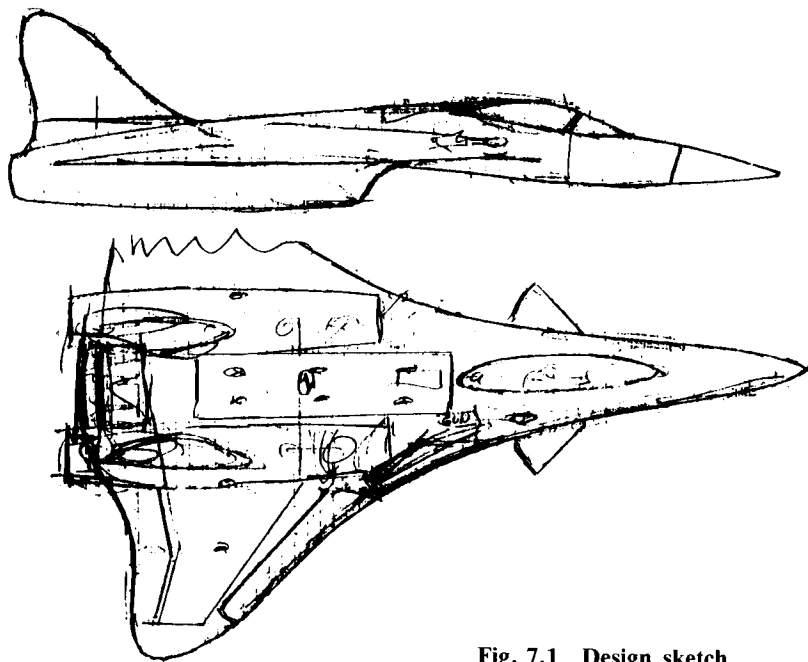


Fig. 7.1 Design sketch.

tem, fuel tanks, and any unique internal components such as a large radar. Conceptual sketches are not usually shown to anybody after the actual layout is developed, but may be used among the design engineers to discuss novel ideas before they begin the layout.

The actual design layout is developed using the techniques to be discussed in the following chapters. Figure 7.2 shows such a design layout, Rockwell's entry in the competition to build the X-29 Forward Sweep Demonstrator. This drawing typifies initial design layouts developed by major airframe companies during design studies.

Figure 7.3 is the initial design developed from the sketch shown as Fig. 7.1. In this case a computer-aided conceptual design system was used to develop a three-dimensional geometric model of the aircraft concept (Ref. 14). The design techniques are similar whether a computer or a drafting board is used for the initial design.

A design layout such as those shown in Figs. 7.2 and 7.3 represents the primary input into the analysis and optimization tasks discussed in Chapters 12–19. Three other inputs must be prepared by the designer: the wetted-area plot (Fig. 7.4), volume distribution plot (Fig. 7.5), and fuel-volume plots for the fuel tanks. Preparation of the wetted-area and volume plots is discussed later in this chapter; the fuel-volume determination is discussed in Chapter 10.

Once the design has been analyzed, optimized, and redrawn for a number of iterations of the conceptual design process, a more detailed drawing can be prepared. Called the "inboard profile" drawing, this depicts in much

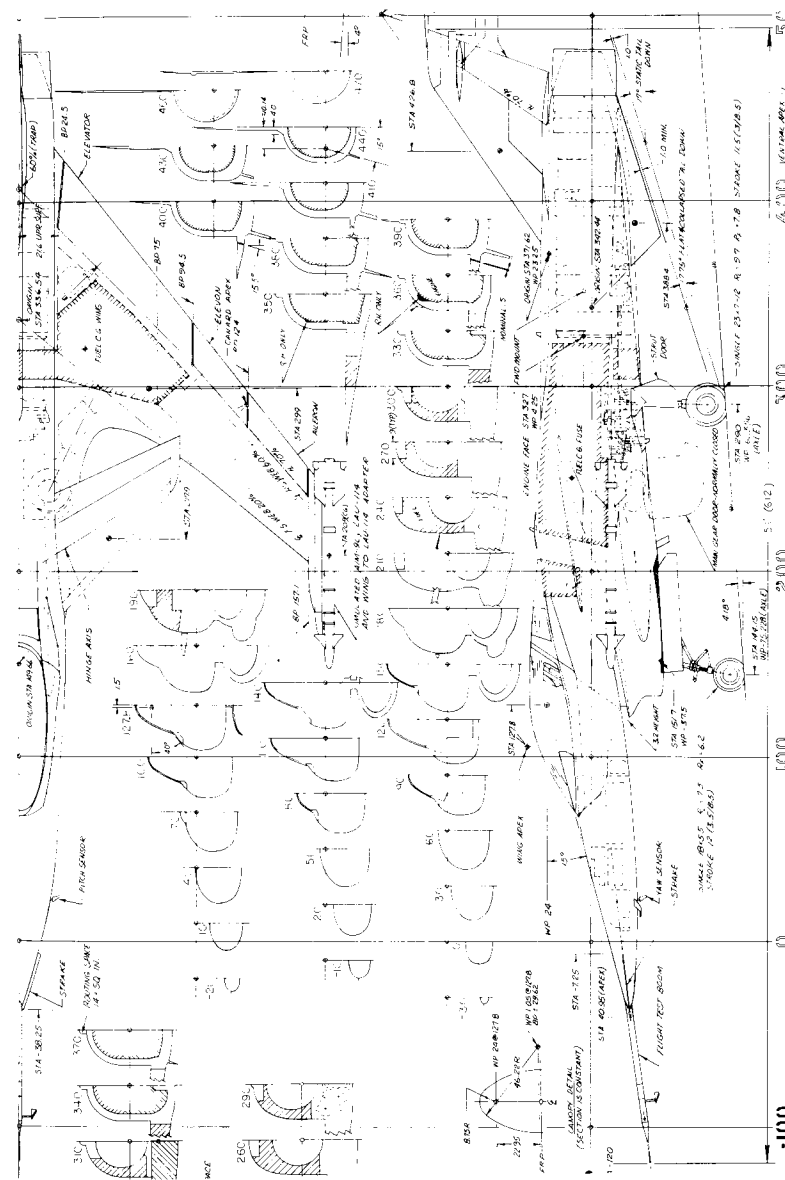


Fig. 7.2 FSW design layout.

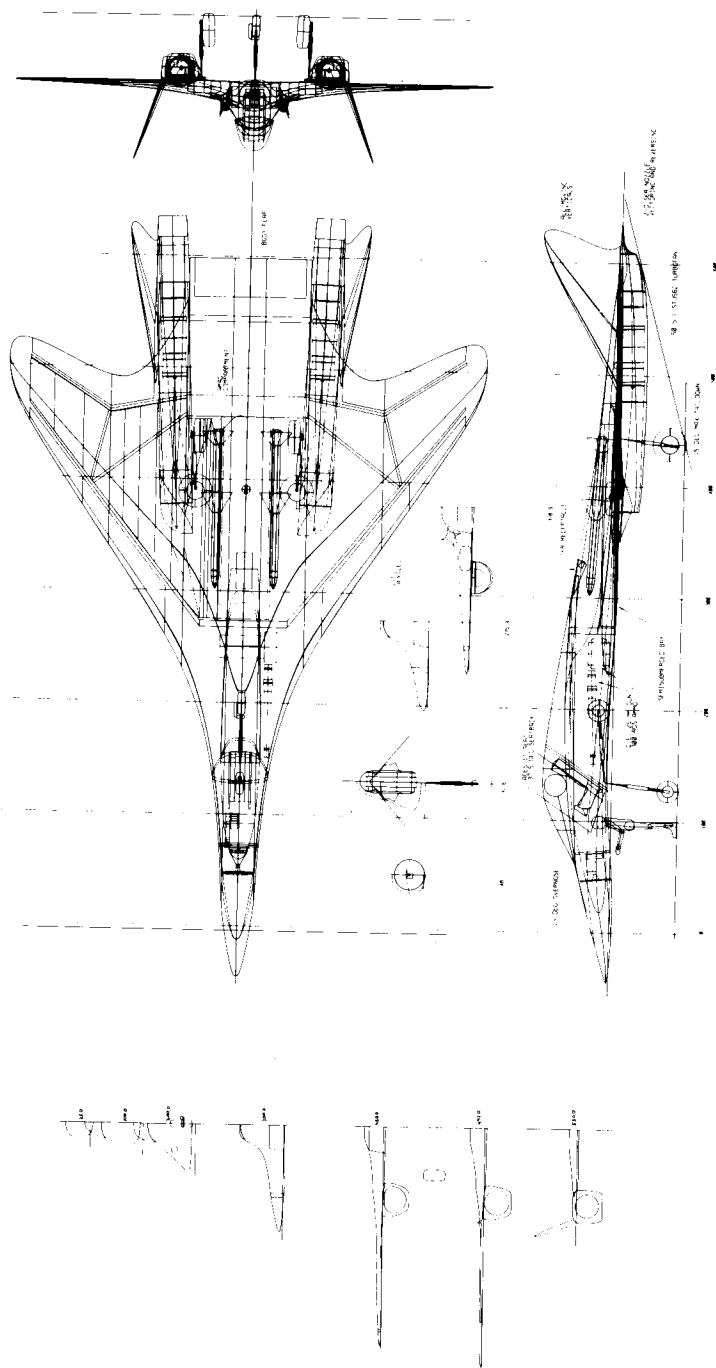


Fig. 7.3 Design layout on a CAD system.

COMPONENT	SURFACE
FUSELAGE	70344.8
VERT TAIL	26165.3
WING	102636.7
CIRCULAR ARC CANOPY	9071.4
NACELLE	25462.9
TOTAL	233681.0

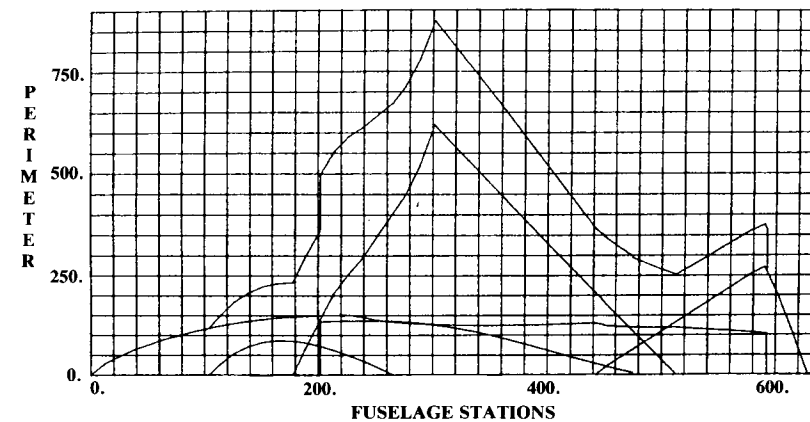


Fig. 7.4 Wetted area plot.

COMPONENT	VOLUME
FUSELAGE	847124.4
VERT TAIL	42903.5
WING	287005.5
CIRCULAR ARC CANOPY	46014.0
NACELLE	95149.8
TOTAL	1318196.8

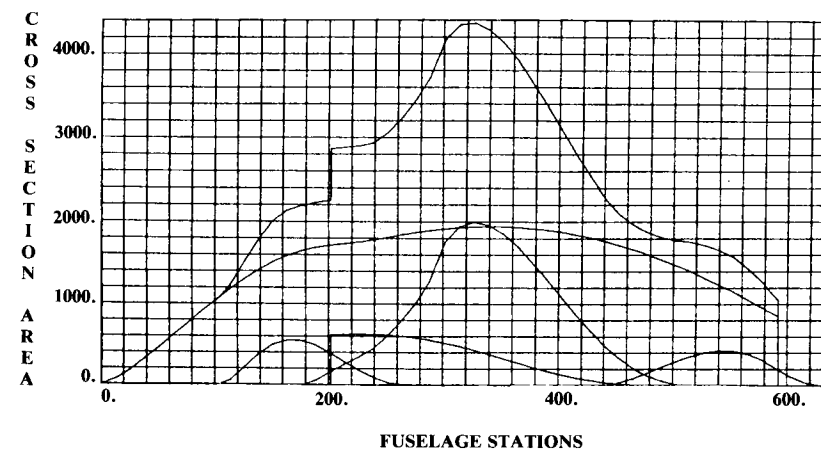


Fig. 7.5 Volume distribution plot.

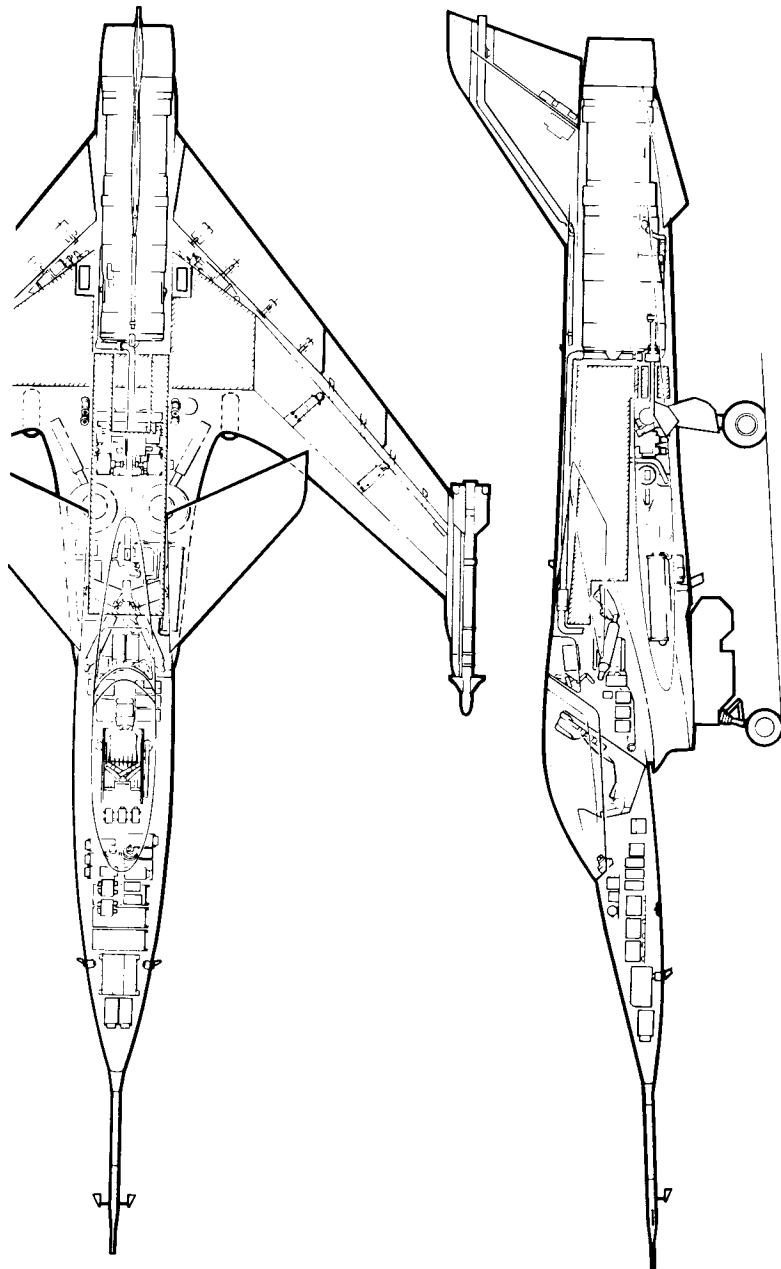


Fig. 7.6 FSW Inboard profile.

greater detail the internal arrangement of the subsystems. Figure 7.6 illustrates the inboard profile prepared for the design of Fig. 7.2. A companion drawing, not shown, would depict the internal arrangement at 20–50 cross-sectional locations.

The inboard profile is far more detailed than the initial layout. For example, while the initial layout may merely indicate an avionics bay based upon a statistical estimate of the required avionics volume, the inboard profile drawing will depict the actual location of every piece of avionics (i.e., “black boxes”) as well as the required wire bundles and cooling ducts.

The inboard profile is generally a team project, and takes many weeks. During the preparation of the inboard profile it is not uncommon to find that the initial layout must be changed to provide enough room for everything. As this can result in weeks of lost effort, it is imperative that the initial layout be as well thought-out as possible.

Figure 7.7 shows a side-view inboard profile prepared in 1942 for an early variant of the P-51. This detailed drawing shows virtually every internal system, including control bellcranks, radio boxes, and fuel lines. Preparation of such a detailed drawing goes beyond the scope of this book, but aspiring designers should be aware of them.

At about the same time that the inboard profile drawing is being prepared, a “lines control” drawing may be prepared that refines and details the external geometry definition provided on the initial layout. Again, such a detailed drawing goes beyond the scope of this book. Also, most major companies now use computer-aided lofting systems that do not require a lines control drawing.

After the inboard profile drawing has been prepared, an “inboard isometric” drawing (Fig. 7.8) may be prepared. It will usually be prepared by the art group for the purpose of illustration only, and be used in briefings and proposals. Such a drawing is frequently prepared and published by aviation magazines for existing aircraft. (In fact, the magazine illustrations are usually better than those prepared by the aircraft companies!)

7.3 CONIC LOFTING

“Lofting” is the process of defining the external geometry of the aircraft. “Production lofting,” the most detailed form of lofting, provides an exact, mathematical definition of the entire aircraft including such minor details as the intake and exhaust ducts for the air conditioning.

A production-loft definition is expected to be accurate to within a few hundredths of an inch (or less) over the entire aircraft. This allows the different parts of the aircraft to be designed and fabricated at different plant sites yet fit together perfectly during final assembly. Most aircraft companies now use computer-aided loft systems that incorporate methods discussed in Ref. 80.

For an initial layout it is not necessary to go into as much detail. However, the overall lofting of the fuselage, wing, tails, and nacelles must be defined sufficiently to show that these major components will properly enclose the required internal components and fuel tanks while providing a smooth aerodynamic contour.

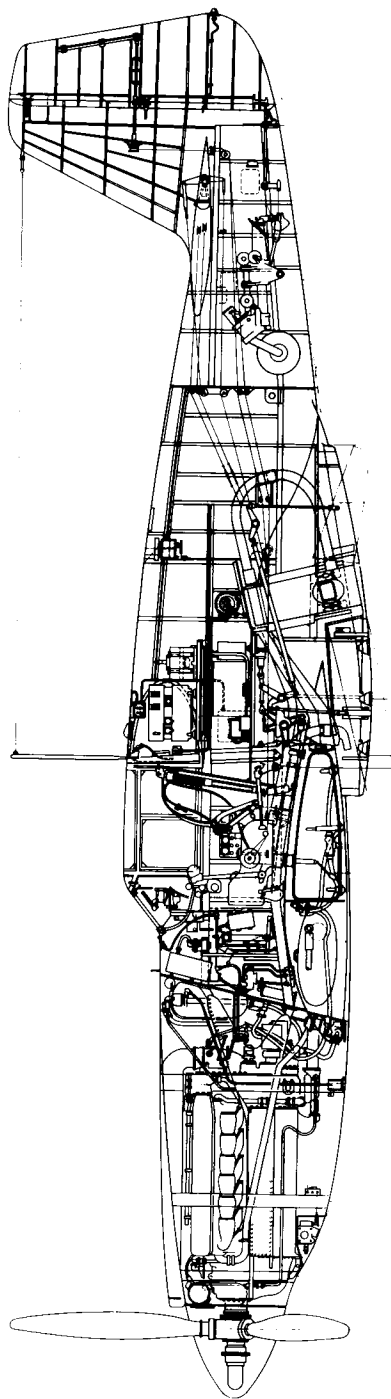


Fig. 7.7 P-51 variant inboard profile.

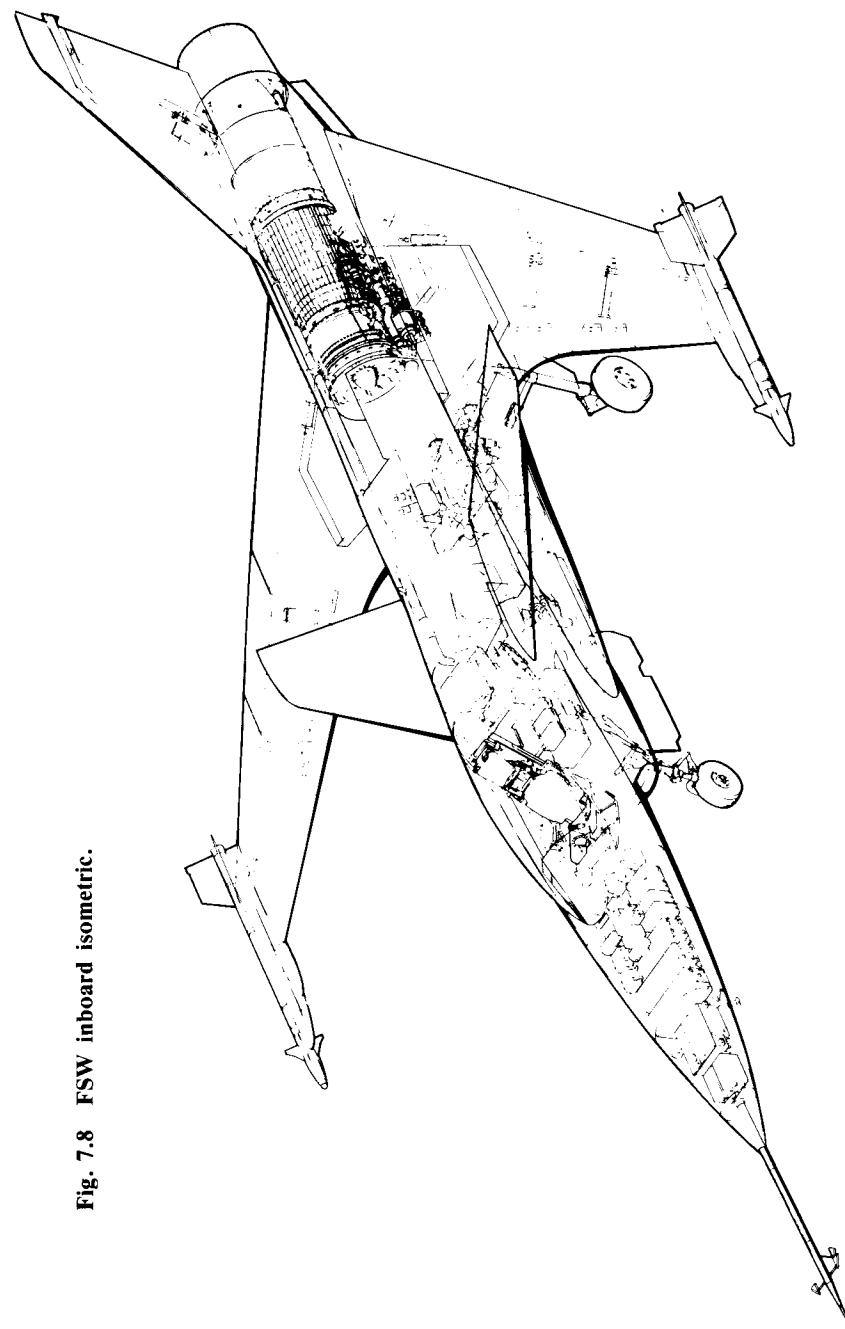


Fig. 7.8 FSW inboard isometric.

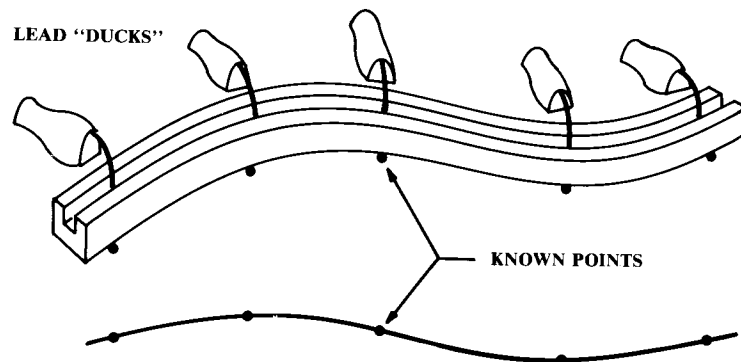


Fig. 7.9 Spline lofting.

Lofting gets its name from shipbuilding. The definition of the hull shape was done in the loft over the shipyard, using enormous drawings. To provide a smooth longitudinal contour, points taken from the desired cross sections were connected longitudinally on the drawing by flexible "splines," long, thin wood or plastic rulers held down at certain points by lead "ducks" (pointed weights—see Fig. 7.9).

This technique was used for early aircraft lofting, but suffers from two disadvantages. First, it requires a lot of trial and error to achieve a smooth surface both in cross section and longitudinally.

Second, and perhaps more important, this method does not provide a unique mathematical definition of the surface. To create a new cross section requires a tremendous amount of drafting effort, especially for a canted cross section (i.e., a cross-sectional cut at some angle other than perpendicular to the centerline of the aircraft). In addition to the time involved, this method is prone to mismatch errors.

A new method of lofting was used for the first time on the P-51 Mustang (Ref. 15). This method, now considered traditional, is based upon a mathematical curve form known as the "conic."

The great advantage of the conic is the wide variety of curves that it can represent, and the ease with which it can be constructed on the drafting table.

While many other forms of lofting are in use, conic lofting has been the most widely used. Also, an understanding of conic lofting provides the necessary foundation to learn the other forms of lofting, including computer-aided lofting.

A conic is a second-degree curve whose family includes the circle, ellipse, parabola, and hyperbola. The generalized form of the conic is given in Eq. (7.1). The conic is best visualized as a slanted cut through a right circular cone (Fig. 7.10). A number of specialized conic equations are provided in Ref. 80.

$$C_1X^2 + C_2XY + C_3Y^2 + C_4X + C_5Y + C_6 = 0 \quad (7.1)$$

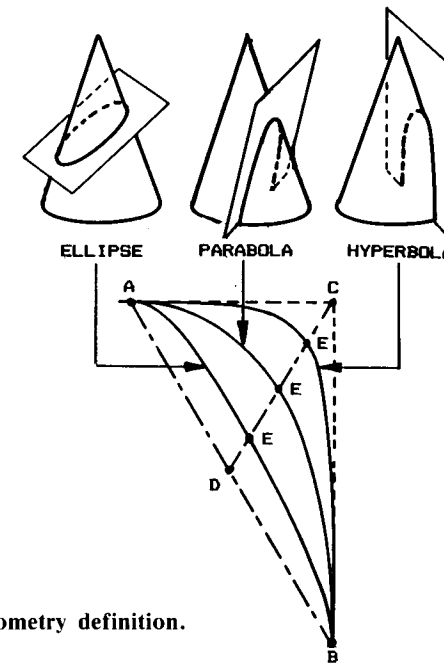


Fig. 7.10 Conic geometry definition.

The shape of the conic depends upon the angle of the cut through the cone. If the cut is flat (i.e., perpendicular to the axis of the cone), then the resulting curve will be a circle; if somewhat slanted, an ellipse; if exactly parallel to the opposite side, a parabola. A greater cut angle yields a hyperbola.

A conic curve is constructed from the desired start and end points ("A" and "B"), and the desired tangent angles at those points. These tangent angles intersect at point "C." The shape of the conic between the points A and B is defined by some shoulder point "S." (The points labeled "E" in Fig. 7.10 are a special type of shoulder point, discussed later.) Figure 7.11 illustrates the rapid graphical layout of a conic curve.

The first illustration in Fig. 7.11 shows the given points A, B, C, and S. In the second illustration, lines have been drawn from A and B, passing through S.

The remaining illustrations show the generation of one point on the conic. In the third illustration a line is drawn from point C at an arbitrary angle. Note the points where this line intersects the A-S and B-S lines.

Lines are now drawn from A and B through the points found in the last step. The intersection of these lines is a point "P" which is on the desired conic curve.

To generate additional points, the last two steps are repeated. Another line is drawn from point C at another arbitrary angle, and then the lines from A and B are drawn and their intersection is found. When enough points have been generated, a French curve is used to draw the conic.

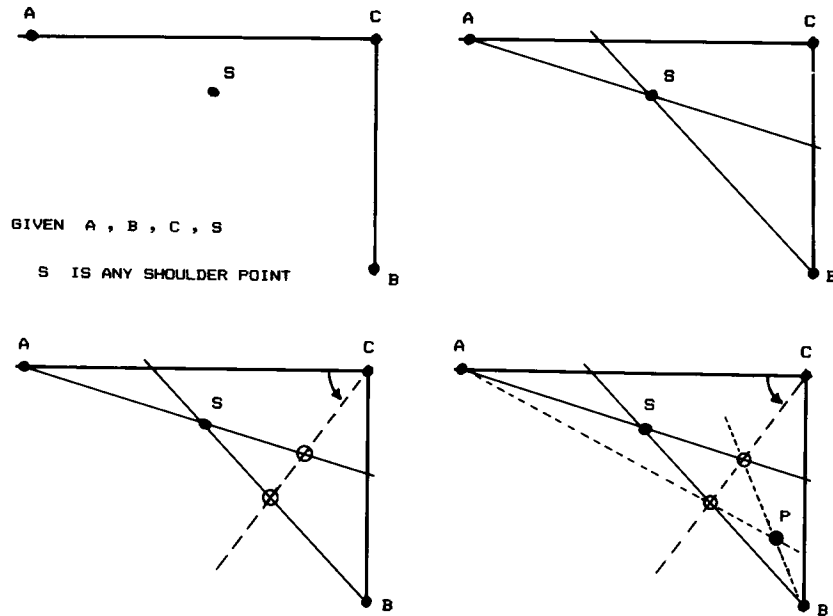
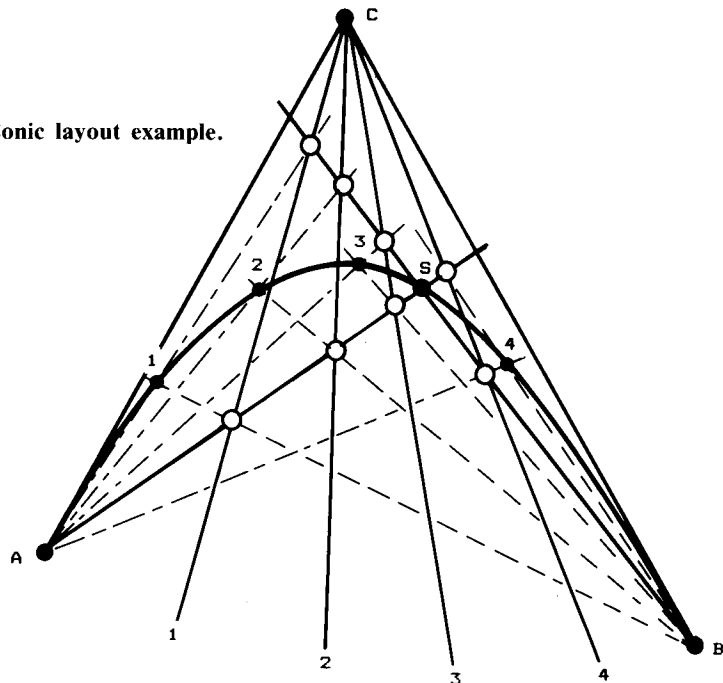


Fig. 7.11 Conic layout.

Fig. 7.12 Conic layout example.



While this procedure seems complicated at first, with a little practice a good designer can construct an accurate conic in less than a minute. Figure 7.12 illustrates a conic curve generated in this manner. Note that it is not necessary to draw completely the various lines, as it is only their intersections which are of interest.

7.4 CONIC FUSELAGE DEVELOPMENT

Longitudinal Control Lines

To create a smoothly-lofted fuselage using conics, it is necessary only to ensure that the points A , B , C , and S in each of the various cross sections can be connected longitudinally by a smooth line. Figure 7.13 shows the upper half of a simple fuselage, in which the A , B , C , and S points in three cross sections are connected by smooth longitudinal lines. These are called "longitudinal control lines" because they control the shapes of the conic cross sections.

Figure 7.14 shows the side and top views of these longitudinal control lines. Since the cross sections are tangent to horizontal at the top of the fuselage, the A and C lines are identical in side view. Similarly, the cross sections are tangent to vertical at the side of the fuselage, so the B and C lines are identical in top view. This is common, but not required.

In Fig. 7.14, the longitudinal control lines are used to create a new cross section, in between the second and third cross sections previously defined. This new cross section is created by measuring, from the longitudinal control lines, the positions of the A , B , C , and S points at the desired location of the new cross section.

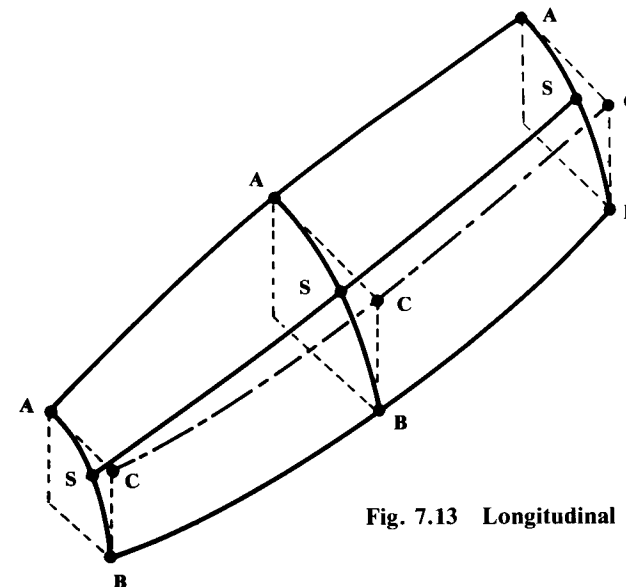


Fig. 7.13 Longitudinal control lines.

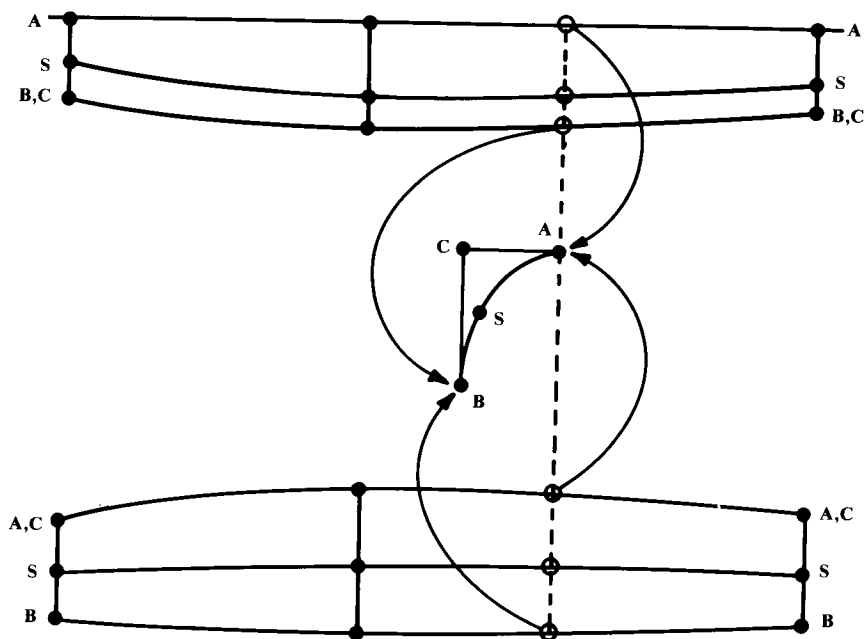


Fig. 7.14 Cross section development from longitudinal control lines.

As is shown for point *A*, each point is defined by two measurements, one from side view and one from top view. From these points the new cross section can be drawn using the conic layout procedure illustrated in Fig. 7.11.

The original cross sections that are used to develop the longitudinal control lines are called the “control cross sections,” or “control stations.” These cross sections are drawn to enclose the various internal components, such as the cockpit or engine.

Control stations can also be drawn to match some required shape. For example, the last cross section of a single-engined jet fighter with a conventional round nozzle would have to be a circle of the diameter of the nozzle.

Typically, some five to ten control stations will be required to develop a fuselage that meets all geometric requirements. The remaining cross sections of the fuselage can then be drawn from the longitudinal control lines developed from these control stations.

Fuselage Lofting Example

Figure 7.15 illustrates a common application of conic lofting to define a fighter fuselage for an initial layout. Five control stations are required for this example. Station 0 is the nose, which is a single point. All the longitudinal control lines must originate there.

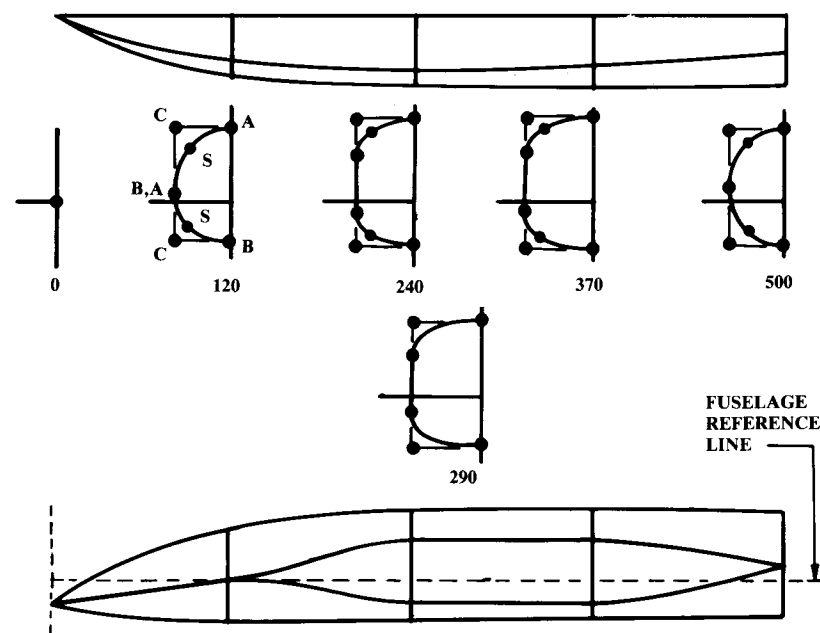


Fig. 7.15 Typical fuselage lofting.

Station 120 is established for this example by the requirements for the cockpit (Chapter 9). This station is approximately circular in shape, and is defined using two conics (upper and lower). Each conic has its own *A*, *B*, *C*, and *S* points. Note that the *B* (end) point of the upper conic is identical to the *A* (start) point of the lower conic.

Station 240 has a flat side to provide for a side-mounted inlet as can be seen on the F-4, the MiG-23, and many other aircraft. At this station, the end points of the upper and lower conics are moved apart vertically, with the area between them defined as a straight line. Note in side view that the longitudinal control lines separate smoothly, not suddenly. This is to ensure a smooth longitudinal contour.

Station 370 is similar to station 240, with a relatively square cross-sectional shape. This could allow room for the landing gear or perhaps to attach a low wing to the side of the fuselage, without a drag-producing acute angle.

Station 500 is a circular cross section, to allow for a connection with a round exhaust nozzle. The longitudinal control lines come back together in a smooth fashion, as shown.

These five control stations are then used to create the longitudinal control lines. From those lines, additional cross sections can be created as desired. Section 290 was created in such a fashion, by measuring the conic control points from the longitudinal control lines and then drawing the conics as previously described.

Figure 7.15 shows only the fuselage lofting. The canopy, inlet duct, and inlet duct fairing would be lofted in a similar fashion, using longitudinal control lines through a few control stations.

Conic Shape Parameter

One problem arises with this method of initial lofting. The locations of the shoulder points (S) can be difficult to control, creating conics either too square (shoulder point too close to point C) or too flat (shoulder point too far away from point C). An alternate technique using conics involves a parameter which directly controls the shoulder point's distance from the point C .

The points labeled E in Fig. 7.10 are conic shoulder points which happen to lie upon the line $D-C$. " D " is the point exactly midway between A and B . Such a shoulder point E determines the "conic shape parameter (ρ)," as defined in the following equation:

$$\rho = |\overline{DE}| / |\overline{DC}| \quad (7.2)$$

where

$$|\overline{AD}| = |\overline{BD}| \quad (7.3)$$

Referring to Fig. 7.10, the shoulder points labeled E are based upon the ρ values required to obtain the ellipse, parabola, or hyperbola forms of the conic. These are given below, along with the ρ value that defines a circle (a special form of the ellipse):

$$\begin{aligned} \text{Hyperbola: } & \rho > 0.5 \\ \text{Parabola: } & \rho = 0.5 \\ \text{Ellipse: } & \rho < 0.5 \\ \text{Circle: } & \rho = 0.4142 \quad \text{and} \quad |\overline{AC}| = |\overline{BC}| \end{aligned} \quad (7.4)$$

The conic shape parameter allows the designer to specify the conic curve's distance from the point C . A conic with a large ρ value (approaching 1.0) will be nearly square, with the shoulder point almost touching the point C . A conic with a small ρ value (approaching 0.0) will nearly resemble the straight line from $A-B$. The parameter ρ can be used to control more easily the longitudinal fairing of a fuselage.

Figure 7.16 shows the use of the conic shape parameter (ρ) to lay out a conic. Points A , B , and C are known, but the shoulder point S is not known. However, the value of ρ is given.

In the illustration on the right side of Fig. 7.16, the line $A-B$ has been drawn and bisected to find the point D . The shoulder point S is found by measuring along line $D-C$, starting at D , by a distance equal to ρ times the total length of line $D-C$. Once the shoulder point is found, the conic can be drawn as illustrated in Fig. 7.11.

By using this approach, a fuselage can be lofted without the use of a longitudinal control line to control the location of the shoulder points. If ρ is specified to be some constant value for all of the cross sections, then the

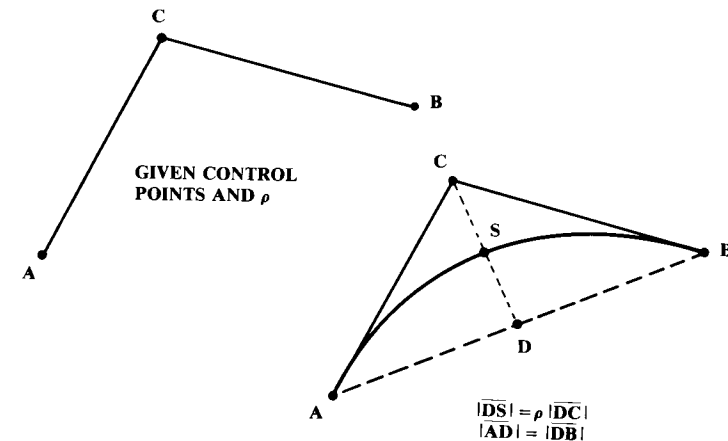


Fig. 7.16 Conic layout using ρ .

designer need only control the conic endpoints and tangent intersection points. To permit the fuselage ends to be circular in shape, the value of ρ would be fixed at 0.4142.

Greater flexibility can be attained by allowing ρ to vary longitudinally. For example, the fuselage of Fig. 7.15 requires a ρ value of 0.4142 at both ends to allow a circular shape, but the values of ρ at the middle of the fuselage are higher, perhaps around 0.7.

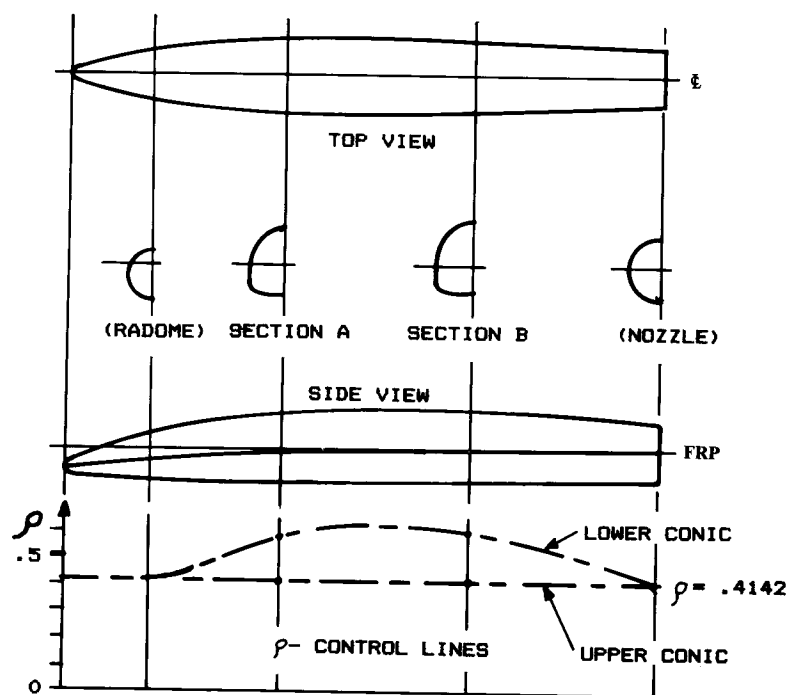
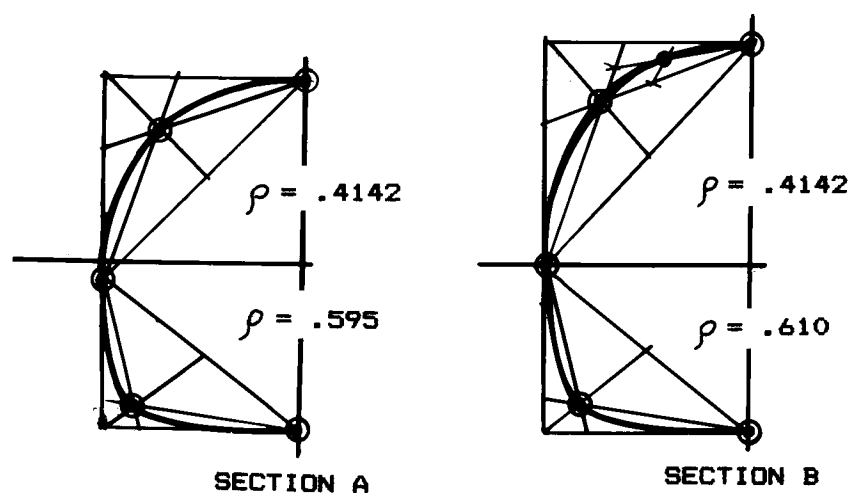
An "auxiliary control line" can be used to control graphically the value of ρ , as shown in Fig. 7.17. Note the auxiliary control line for ρ at the bottom. If the value of ρ varies smoothly from nose to tail, and the conic endpoints and tangent intersection point are controlled with smooth longitudinal lines, then the resulting fuselage surface will be smooth.

In Fig. 7.17 the upper conic has a constant ρ value of 0.4142, while the lower conic has a ρ value varying from 0.4142 at the nose and tail to about 0.6 at the middle of the fuselage. This has the effect of "squaring" the lower fuselage to provide more room for the landing gear.

Figure 7.18 shows the use of ρ to develop the cross sections labeled A and B . Observe the development of the upper and lower conics by the method shown previously in Fig. 7.16, and the use of different ρ values for the upper and lower conics.

Thus far, no mention has been made of the method for developing the longitudinal control lines and auxiliary control lines. During production lofting, these control lines would be defined mathematically, using conics or some form of polynomial.

For initial layouts, sufficient accuracy can be obtained graphically through the use of the flexible splines discussed earlier. Points are taken from the control cross sections and plotted in side and top view, then connected longitudinally using a spline to draft a smooth line. In fact, a designer with a "good eye" can obtain sufficient smoothness using a French curve if spline and ducks are not available.

Fig. 7.17 Conic fuselage development using ρ .Fig. 7.18 Cross section development using ρ .

7.5 FLAT-WRAP FUSELAGE LOFTING

An important cost driver for aircraft fabrication is the amount of compound-curvature used in lofting the aircraft. Compound-curvature implies the existence of surface curvature in all directions for some point on the surface.

For example, a ball is entirely composed of compound-curvature surfaces. A flat sheet has no curvature, compound or otherwise. A cylinder is curved, but only in one direction, so it does not have any compound curvature. Instead, a cylinder or any other surface with curvature in only one direction is said to be "flat-wrapped".

If a surface is flat-wrapped, it can be constructed by "wrapping" a flat sheet around its cross sections. For aircraft fabrication, this allows the skins to be cut from flat sheets and bent to the desired skin contours.

This is far cheaper than the construction technique for a surface with compound curvature. Compound curvature requires that the skins be shaped by a stretching or stamping operation, which entails expensive tools and extra fabrication steps.

Aircraft applications of flat-wrap lofting must be defined in the initial loft definition used for the conceptual layout. There are several ways of lofting a surface so that it is flat-wrapped. The simplest technique uses a constant cross section. For example, a commercial airliner usually has the identical circular-cross-sectional shape over most of its length. In fact, any cross section shape will produce a flat-wrap surface if it is held constant in the longitudinal direction.

Often an identical cross-sectional shape will not be desired, yet a flat-wrap lofting may be attained. If the same cross-sectional shape is maintained but linearly scaled in size, a flat-wrap contour is produced. For example, a cone is a flat-wrap surface produced by linearly scaling a circular cross section.

Many aircraft have a tailcone which, although not circular in cross section, is linearly scaled to produce a flat-wrap surface. This can be accomplished with conics by maintaining identical tangent angles and ρ value, using straight longitudinal control lines, and maintaining the lengths AC and BC in constant proportion.

Sometimes it is necessary to vary the shape of the cross sections other than by scaling. Flat wrap cannot be exactly maintained in such cases using conics. A more sophisticated technique (beyond the scope of this book) must be used.

However, flat wrap can be closely approximated in most such cases on two conditions. First, the longitudinal control lines must be straight. This includes the line controlling the shoulder point (S). If the conic shape parameter (ρ) is used instead of a shoulder-point control line, then the ρ value must be either constant or linearly varied.

Second, the tangent angles of the conics must not change longitudinally. If the tangent angles are all either horizontal or vertical, as in Figs. 7.15 and 7.17, this condition can easily be met.

Figure 7.19 shows such a complex flat-wrapped surface. The fuselage is defined by five conics plus a straight-line, flat underside. The "bump" on top could represent the back of the canopy, and grows smaller towards the

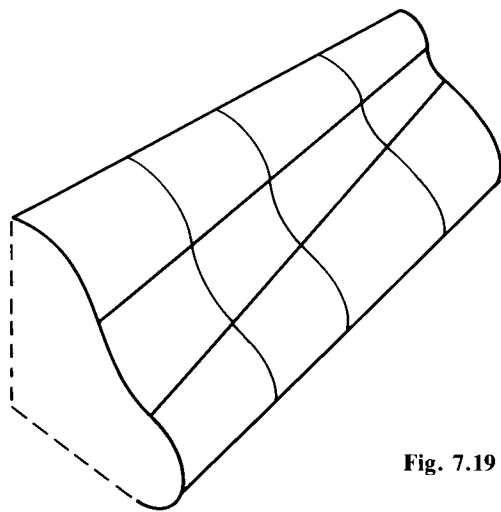


Fig. 7.19 Complex flat-wrapped surface.

rear of the fuselage. While the conics change shape and size, their endpoints hold the same tangent angles.

It is important to realize that the use of flat-wrap lofting for a fuselage represents a compromise. While flat-wrap surfaces are easier and cheaper to fabricate, they are less desirable from an aerodynamic viewpoint. For example, a smoothly contoured teardrop shape will have less drag than a flat-wrap cylinder with a nosecone and tailcone.

7.6 CIRCLE-TO-SQUARE ADAPTER

A common problem in lofting is the “circle-to-square adapter.” For example, the inlet duct of many supersonic jet aircraft is approximately square at the air inlet, yet must attain a circular shape at the engine front-face. Modern, two-dimensional nozzles also require a circle-to-square adapter.

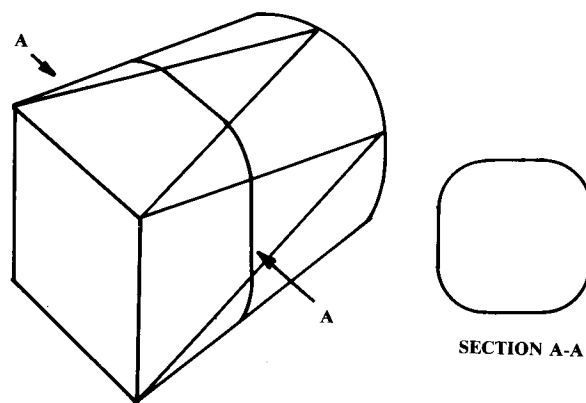


Fig. 7.20 Circle-to-square adapter.

Flat-wrap can be attained for a circle-to-square adapter by constructing the adapter of interlocking, V-shaped segments, each of which is itself flat-wrapped (Fig. 7.20).

The flat sides of the square section taper to points that just touch the circular section. Similarly, the cone-shaped sides of the circular section taper to points that touch the corners of the square section. Note the “rounded-off square” shape of the intermediate sections. The connecting surfaces must be straight longitudinally for a flat-wrap surface to be maintained.

7.7 FUSELAGE LOFT VERIFICATION

The use of smooth longitudinal control lines defining conic cross sections assures a smooth fuselage; but sometimes it is necessary to deviate from this type of definition. For example, if a part of the fuselage is to be flat-wrapped, it may be difficult to smoothly connect the straight control lines for the flat-wrap part of the fuselage with the curved control lines for the rest of the fuselage.

Also, it may be desirable to have two different flat-wrap parts of the fuselage that are directly connected, resulting in an unavoidable break in the smoothness of the longitudinal control lines. In such cases the designer should evaluate the resulting contours to ensure that any breaks in the longitudinal smoothness are not too extreme.

Sometimes a designer will be asked to evaluate a design created by someone else. This is common practice at government agencies such as the Air Force Aeronautical Systems Division and the Naval Air Systems Command. In addition to the analytical evaluations for performance and range, the designer should evaluate the design layout to ensure that the cross sections shown are in fact smooth longitudinally.

These smoothness evaluations are performed using a technique borrowed from shipbuilding. Hull contours are evaluated for smoothness by laying out the “waterlines.” If a ship is floating in the water, the line around the hull where the surface of the water intersects the hull is a waterline. For good ship performance, this waterline should be smooth in the longitudinal direction.

If the hull is raised partly out of the water some arbitrary distance, a new waterline is formed. Hull designers check for hull smoothness by laying out a large number of these waterlines, each separated in height by some arbitrary distance. If all the waterlines have smooth contours, then the hull is smooth.

Such horizontal waterline cuts can be used for evaluation of the smoothness of an aircraft fuselage; but it is more common to use vertically-oriented cuts known as “butt-plane cuts” (Fig. 7.21).

Butt-plane (“butt-plane”) cuts form the intersection of the aircraft with vertical planes defined by their distance from the aircraft centerline. For example, “butt-plane 30” is the contour created by intersecting a vertical plane with the fuselage at a distance of 30 in. from the centerline.

Note in Fig. 7.21 that the butt-plane cuts are oriented such that the airfoil is a butt-plane cut of the wing. It is for this reason that butt-plane cuts are more commonly used for aircraft than waterlines.

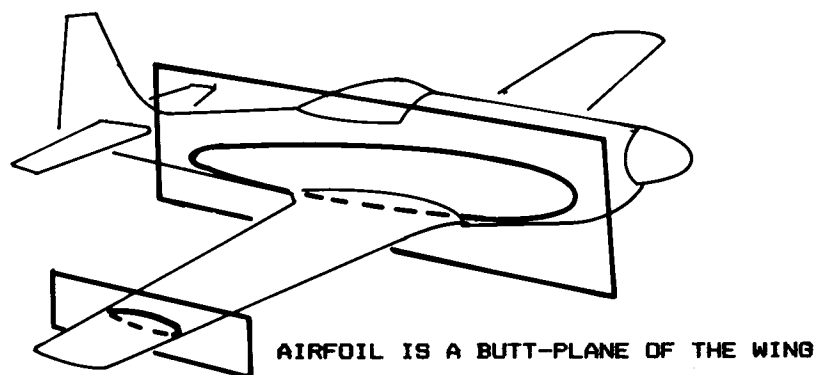


Fig. 7.21 Butt-plate cut.

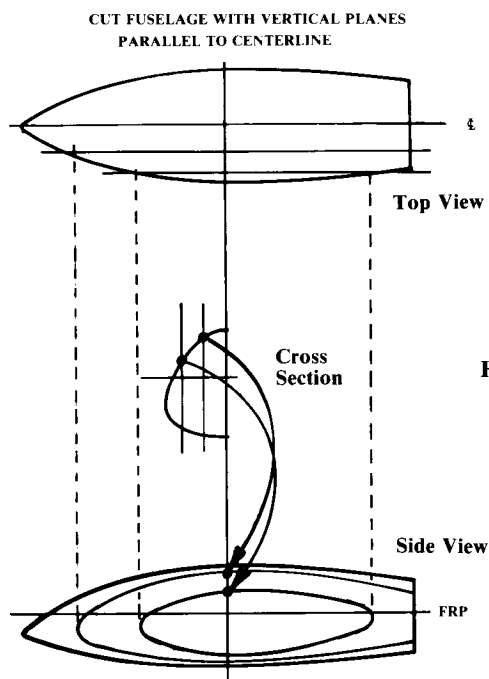


Fig. 7.22 Butt-plate cut layout.

Figure 7.22 illustrates the development of butt-plane cuts. Vertical lines are drawn on each cross section, indicating the locations of the arbitrarily-selected butt-planes. The points where these vertical lines intersect the cross sections are transferred to the side-view drawing and connected longitudinally. If the fuselage surface is smooth, then these longitudinal lines for the different butt-planes will all be smooth.

Buttock-plane cuts can also be used to generate new cross sections. Once the butt-plane cuts are developed as in Fig. 7.22, a new cross section can be

developed by transferring the vertical locations of the butt-plane cuts to the cross section desired, and then drawing a smooth cross-sectional contour using those points.

Sometimes this method is easier than developing the longitudinal control lines for conic fuselage lofting. This is most likely when the surface is highly irregular, such as the forebody of a blended wing-body aircraft like the B-1B.

7.8 WING/TAIL LAYOUT AND LOFT

Reference Wing/Tail Layout

Chapter 4 described the selection of the basic geometric parameters for the wing and tails. These parameters include the aspect ratio (A), taper ratio (λ), sweep, dihedral, and thickness. Also, the selection of an appropriate airfoil was considered. In Chapter 6, the actual sizes for the wing, tails, and fuselage were defined, based upon an initial estimate for the takeoff gross weight.

From these parameters, the geometric dimensions necessary for layout of the reference (trapezoidal) wing or tail can be obtained, as shown in Fig. 7.23 and defined by the following equations:

$$b = \sqrt{AS} \quad (7.5)$$

$$C_{\text{root}} = \frac{2S}{b(1 + \lambda)} \quad (7.6)$$

$$C_{\text{tip}} = \lambda C_{\text{root}} \quad (7.7)$$

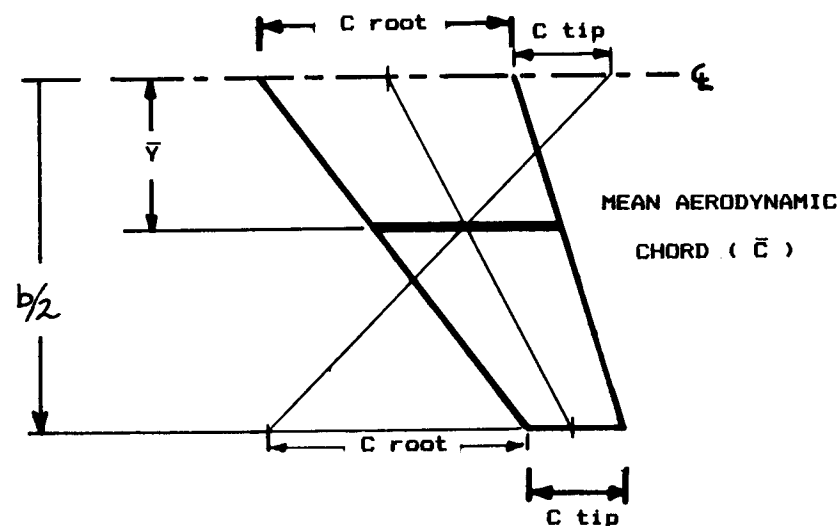


Fig. 7.23 Reference (trapezoidal) wing/tail.

$$\bar{C} = \left(\frac{2}{3}\right) C_{\text{root}} \frac{1 + \lambda + \lambda^2}{1 + \lambda} \quad (7.8)$$

$$\bar{Y} = \left(\frac{b}{6}\right) \left(\frac{1 + 2\lambda}{1 + \lambda}\right) \quad (7.9)$$

For a vertical tail, $\bar{Y}$ is twice the value calculated in Eq. (7.9).

Figure 7.23 also shows a quick graphical method of determining the spanwise (Y) location of the mean aerodynamic chord (MAC or $\bar{C}$), which is mathematically obtained by Eq. (7.9). The location of the mean chord is obtained graphically as the intersection of the 50%-chord line and a line drawn from a point located at the tip chord length behind the root chord to a point at the root chord length ahead of the tip chord.

Wing Location with Respect to the Fuselage

The location and length of the MAC is important because the wing is located on the aircraft so that some selected percent of the MAC is aligned with the aircraft center of gravity. This provides a first estimate of the wing position to attain the required stability characteristics.

For a stable aircraft with an aft tail, the wing should be initially located such that the aircraft center of gravity is at about 30% of the mean aerodynamic chord. When the effects of the fuselage and tail are considered, this will cause the center of gravity to be at about 25% of the total subsonic aerodynamic center of the aircraft.

For an unstable aircraft with an aft tail, the location of the wing depends upon the selected level of instability, but will usually be such that the center of gravity is at about 40% of the mean aerodynamic chord.

For a canard aircraft, such rules of thumb are far less reliable due to the canard downwash and its influence upon the wing. For a control-type canard with a computerized flight control system (i.e., unstable aircraft), the wing can be initially placed such that the aircraft center of gravity is at about 15–25% of the wing's mean aerodynamic chord.

For a lifting-type canard, the mean aerodynamic chords of the wing and canard should both be determined, and the appropriate percent MAC for each should be identified. Then the combined MAC location can be determined as the weighted average of the percent MAC locations for the wing and canard (weighted by their respective areas). Note that this is a very crude estimate!

Chapter 15 provides a quick method of approximating the aircraft center of gravity once the locations of the major internal components are known.

After the initial layout is completed and analyzed using the methods of Chapters 12–19, the wing will probably be moved and the tails resized to meet all required stability and control characteristics. Hopefully the initial estimates will be close enough so that major changes will not be needed.

Wing/Tail Lofting

The reference (trapezoidal) wing and tails are positioned with respect to the fuselage using the methods discussed above. During the layout process the actual, exposed wing and tails will be drawn.

The reference wing is defined to the aircraft centerline, and is based upon the projected area (i.e., dihedral does not affect the top view of the reference wing). The actual, exposed wing begins at the side of the fuselage and includes the effect of the dihedral upon the true-view area. The dihedral angle increases the actual wing area equivalent to dividing by the cosine of the dihedral angle.

Also, the actual wing planform may not be trapezoidal. Figure 7.24 illustrates several of the many nontrapezoidal wing variations. A typical rounded wing tip is shown in Fig. 7.24a. This and other wing-tip shapes have already been discussed. The straightened-out trailing edge shown in Fig. 7.24b increases the flap chord and provides increased wing thickness for the landing gear.

Figure 7.24c illustrates a “Leading-Edge Extension (LEX),” which increases lift for combat maneuvering (see Chapter 12). A highly-blended wing/body is shown in Fig. 7.24d, in which the actual wing looks very little like the reference wing. This type of wing is used to minimize the transonic and supersonic shocks.

Once the designer has settled upon the actual wing and tail planforms, their surfaces must be lofted to provide accurate cross sections. These are required to verify that there is sufficient room for the fuel tanks, landing gear, spars, and other internal components. During production design this lofting would be done using conics or some other mathematical surface definition.

For initial design, simpler methods of wing and tail lofting can be used. These rely upon the assumption that the airfoil coordinates themselves are smoothly lofted. This is an excellent assumption, as otherwise the airfoil performance would be poor.

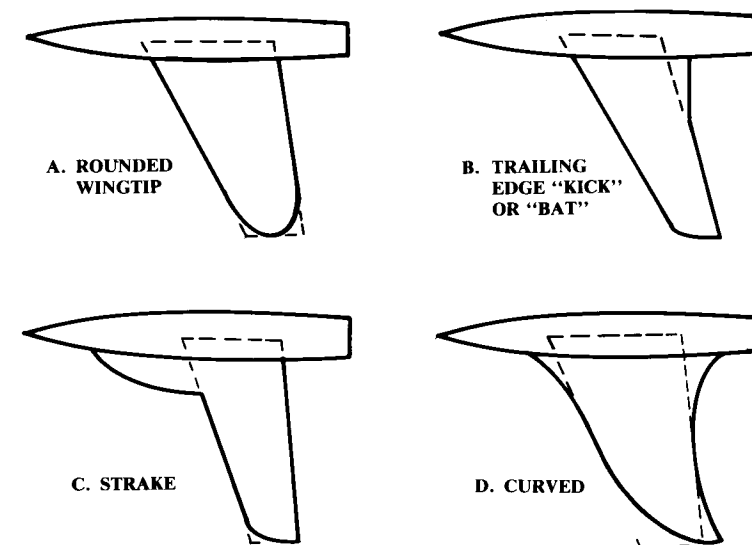


Fig. 7.24 Nontrapezoidal wings.

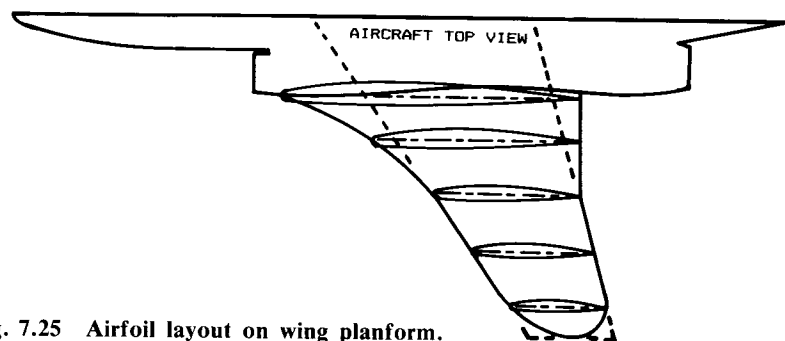


Fig. 7.25 Airfoil layout on wing planform.

If the wing or tail uses the identical airfoil section and thickness ratio at all span stations, and is without twist, the airfoils can be drawn simply by scaling the airfoil coordinates to fit the chord lengths of the selected spanwise locations.

It is customary to lightly draw the airfoils on the top view of the wing, superimposing them on their chordline (Fig. 7.25). This layout procedure simplifies the generation of cross sections, as will be discussed later. For initial design purposes the airfoils can be quickly drawn using only a few scaled coordinate points for the top and bottom surfaces.

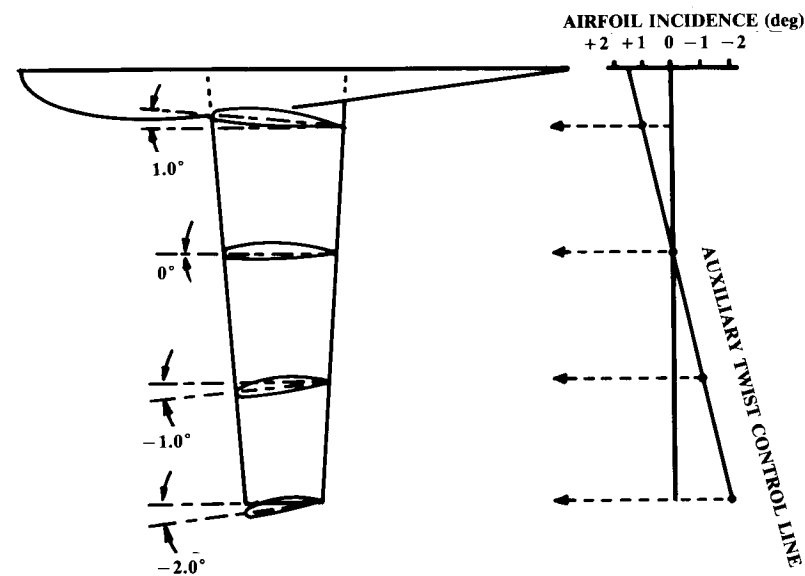
If twist is incorporated, the incidence at each span station must be determined and the chord line rotated accordingly before the airfoil is drawn. Since the chord length is defined in top view, the chord length at each spanwise station must be increased equivalent to dividing by the cosine of the appropriate incidence angle.

Instead of calculating the twist, an auxiliary twist control line may be constructed behind the wing. The airfoil incidence at each span station can then be read from the control line (Fig. 7.26).

A wing with a complicated aerodynamic design may have the twist, camber, and thickness all varying from root to tip. These spanwise variations can be lofted by using a separate auxiliary control line for each, as shown in Fig. 7.27. The airfoil coordinate points must be calculated by separating the airfoil into its camber line and thickness distribution, scaling them as indicated by the auxiliary control lines, and recombining them. Such a complicated wing design is not normally accomplished until much later in the design process.

For a wing such as shown in Fig. 7.27, the complex curvatures of the wing surface may present difficulties. A spar running from root to tip may very well be so curved that it is structurally undesirable. Even worse, the hinge lines for the ailerons and flaps may not lie in a straight line. As curved hinge lines are impossible, the ailerons and flaps may have to be broken into a number of surfaces unless the wing surface can be modified to straighten the hingeline.

This is done by "wing rigging" (not to be confused with the rigging of a biplane wing)—the process of vertically shifting the airfoil sections until some desired spanwise line is straight.



ANGLES ARE EXAGGERATED FOR ILLUSTRATION

Fig. 7.26 Airfoil layout with twist.

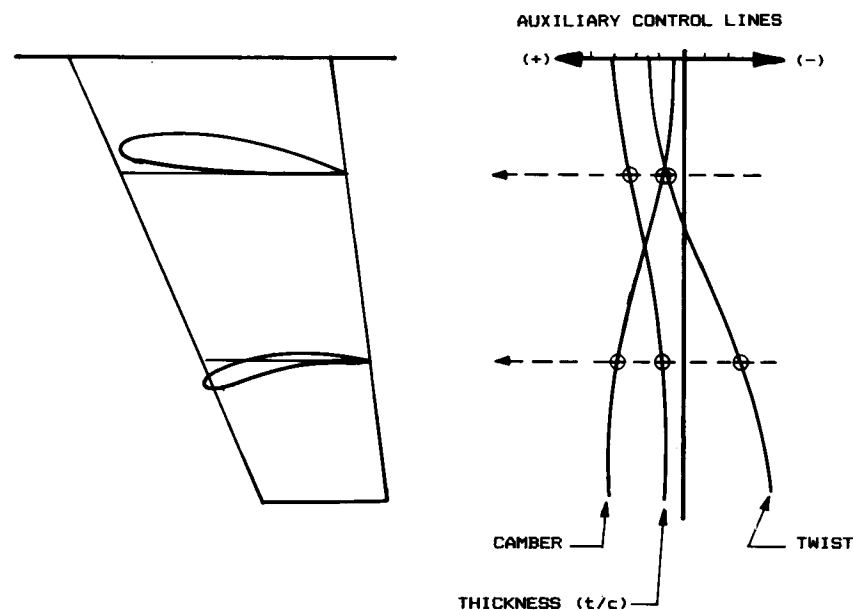


Fig. 7.27 Wing airfoil layout—nonlinear variations.

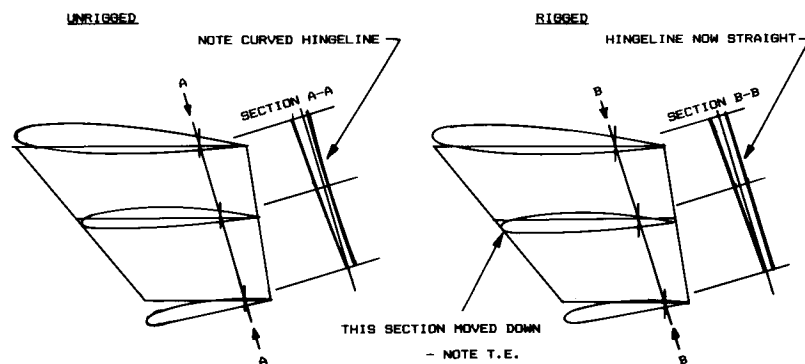


Fig. 7.28 Wing airfoil rigging.

Figure 7.28 illustrates a complex wing in which the aileron hingeline, Section A-A, is curved. On the right side of the figure is the same wing with the midspan airfoil moved downward a few inches. This provides a straight hingeline shown as Section B-B.

Airfoil Linear Interpolation

Most wings are initially defined by a root airfoil and a tip airfoil, which may be different, and their incidence angles or relative twist. Frequently the tip airfoil will be selected for gentle stall characteristics while the root airfoil is selected for best performance. The resulting wing has good overall performance, with good stall characteristics because the tip will stall after the root. The airfoils between the root and tip can be quickly developed by one of two methods.

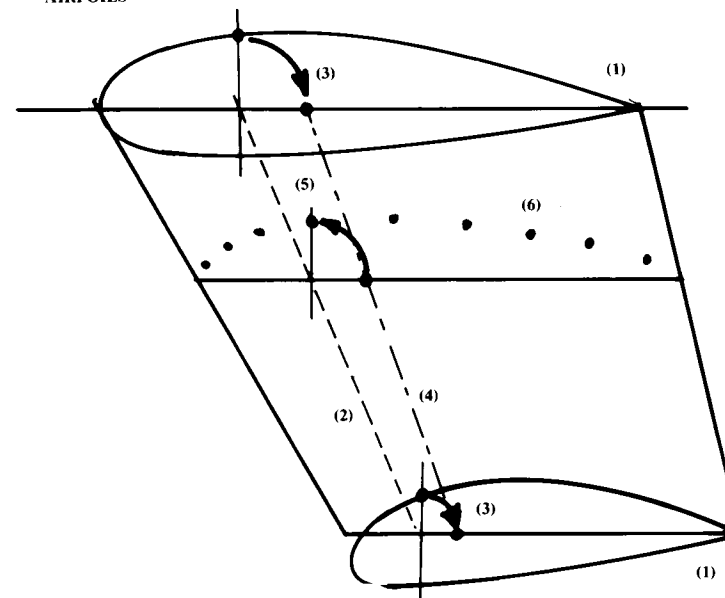
Linear interpolation, the easiest method, is depicted in Fig. 7.29. Here the new airfoils are created as “weighted averages” of the root and tip airfoils. Linearly interpolated airfoils have section properties that are approximately the interpolation of the section properties of the root and tip airfoils.

(Some modern laminar airfoils will not provide interpolated section characteristics. Instead, the interpolated airfoils must each be separately analyzed.)

The intermediate airfoils are linearly interpolated by a five-step process. The root and tip airfoils are drawn (step 1). A constant percent-chord line is drawn connecting the root and tip airfoil, and vertical lines are drawn from the intersection of that line with the chordlines (step 2). The airfoil points found at those vertical lines are “swung down” to the chord line, using an arc centered at the intersection of the chord line and the vertical line (step 3). These “swung down” points for the root and tip airfoils are then connected by a straight line (step 4).

At the desired location of an interpolated airfoil, a chord line is drawn. The intersection of that chord line with the line drawn in step 4 defines the chordwise location of a point on the interpolated airfoil. In step 5 this point is “swung up” to its thickness location by an arc centered at the intersection of the chord line and the spanwise percent-chord line from step 2.

GRAPHIC INTERPOLATION BETWEEN DIFFERING ROOT AND TIP AIRFOILS



- 1 - SUPERIMPOSE ROOT AND TIP AIRFOILS ON PLANFORM
- 2 - DRAW LINE AT SOME CONSTANT PERCENT OF CHORD
- 3 - SWING AIRFOIL POINT DOWN ONTO CHORD REFERENCE LINE
- 4 - CONNECT ROOT AND TIP POINTS FROM 3
- 5 - SWING POINT UP TO NEW AIRFOIL LOCATION
- 6 - REPEAT FOR OTHER PERCENT CHORD LINES

Fig. 7.29 Wing airfoil layout—linear interpolation.

This process is repeated for as many points as are needed to draw the new airfoil. Then the process is repeated to draw other airfoils. While it seems complicated, a wing can be developed using this method in about fifteen minutes by an experienced designer. (However, a computer does this in seconds!)

Airfoil Flat-Wrap Interpolation

The linear-interpolation method doesn't necessarily provide a flat-wrap surface. In laying out a fuselage for flat-wrap, it was necessary to hold the same tangent angle for the conics in the different cross sections. The same is true for wings.

To provide a flat-wrap wing, it is necessary to interpolate between airfoil coordinates with the same slope (i.e., tangent angle). The linear interpolation method connects points based upon their percent of chord. If the wing is twisted or the airfoils are dissimilar, the surface slopes may be different for airfoil points that are at the same percent of chord. This requires a modification to the method described above.

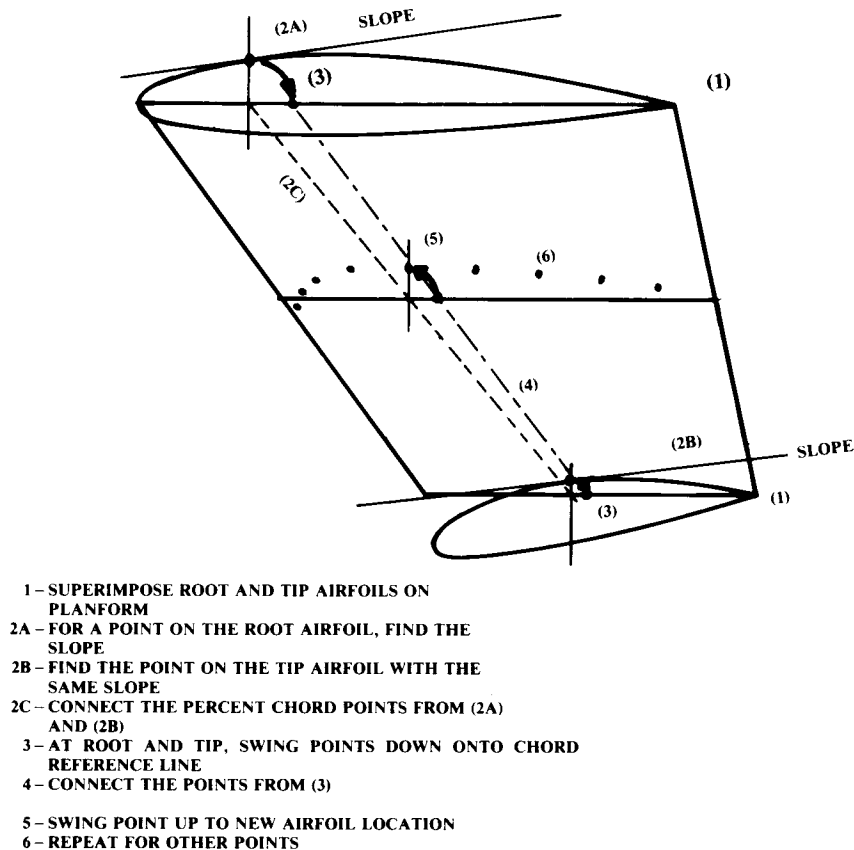


Fig. 7.30 Wing airfoil layout—flat-wrap.

Figure 7.30 illustrates this modification. The only difference is in step 2. Previously a spanwise line was drawn connecting constant percent chord locations on the chordline. To obtain a flat-wrap surface this spanwise line must be drawn connecting locations on the chordline that have the same surface slope. Note in the figure how the tip chord has the indicated slope at a more-aft percent location of the chord than does the root chord.

A number of composite homebuilts are being fabricated by a method long used for model airplanes. A large block of urethane foam is cut directly to the desired wing shape using a hot-wire cutter which is guided by root and tip airfoil templates attached to the foam block. The templates have tic-marks that are numbered. The wire is guided around the templates by two homebuilders, one of whom calls out the numbers of the tic-marks.

If the tic-marks are at constant percent-chord locations, and the wing has dissimilar airfoils or appreciable twist, this method will produce a linearly-interpolated instead of a flat-wrap surface. If the wing is to be covered by

fiberglass, this will pose no problem as the fiberglass cloth will easily conform to the slight amount of compound-curvature present.

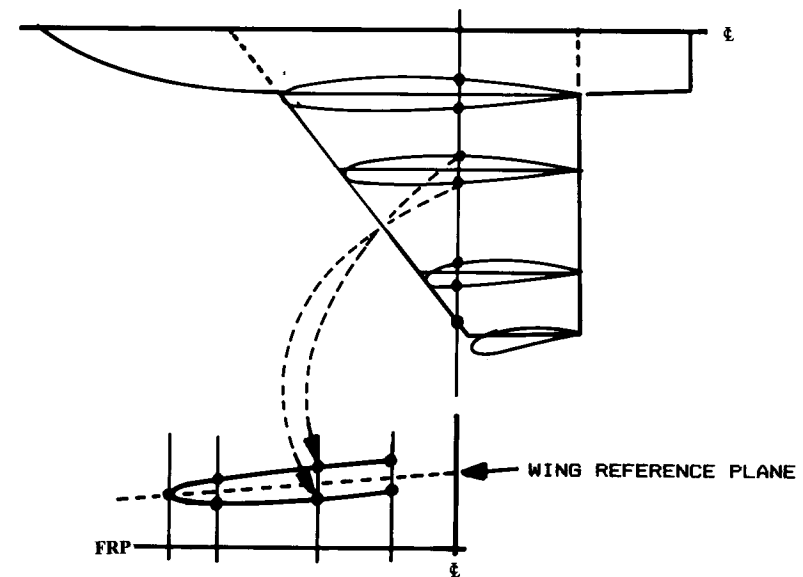
However, if the wing is to be covered by sheet metal or plywood, the linearly-interpolated foam surface will be depressed relative to the flat-wrapped skin. This could reduce the strength of the skin bonding. It is conceivable that such a wing could fail in flight for this simple reason. Who said lofting isn't important?

Wing/Tail Cross-Section Layout

Wing lofting during initial design permits verification that the fuel and other internal components will fit within the wing. This requires the development of wing and tail cross sections oriented perpendicular to the aircraft centerline.

Such cross sections can be easily developed once the airfoils are drawn onto the top view of the wing. Figure 7.31 illustrates the development of one such cross section.

To develop a wing (or tail) cross section, vertical lines are drawn on the cross section at the spanwise locations of the airfoils shown on the wing top view. Also, the wing reference plane is shown at the appropriate wing dihedral angle. Then the airfoil upper and lower points are measured relative to the plane of the wing, and drawn accordingly on the cross section. The cross-section shape can then be drawn using French curves.



SUPERIMPOSE AIRFOILS ON PLANFORM

TRANSFER POINTS TO CROSS SECTION

Fig. 7.31 Wing/tail cross section layout.

The same procedure can be used to develop section cuts at angles other than perpendicular to the aircraft centerline. The sections of Fig. 7.28 labeled A-A and B-B were developed in this manner.

Wing Fillets

For improved aerodynamic efficiency, the wing-fuselage connection of most aircraft is smoothly blended using a "wing fillet" (Fig. 7.32). A wing fillet is generally defined by a circular arc of varying radius, tangent to both the wing and fuselage. Typically a wing fillet has a radius of about 10% of the root-chord length.

The fillet circular arc is perpendicular to the wing surface, so the arc is in a purely vertical plane only at the maximum thickness point of the wing. At the leading edge, the arc is in a horizontal plane.

The fillet arc radius may be constant, or may be varied using an auxiliary radius control line, as shown in Fig. 7.32. Note that the starting radius must be equal to the fillet radius shown in the wing top view. Also, the fillet radius is usually increasing towards the rear of the aircraft, to minimize airflow separation.

Some aircraft have a fillet only on the rear part of the wing. In this case the fillet starts, with zero radius, at the wing's maximum thickness point.

For initial layout purposes the fillet is frequently "eyeballed." Only a few of the 10 or 15 aircraft cross sections developed for an initial layout will show the wing fillet, so a fillet radius that "looks good" can be used. If better accuracy is desired, a fillet radius control line can be used with the simplifying assumption that the fillet arc is approximately vertical over most

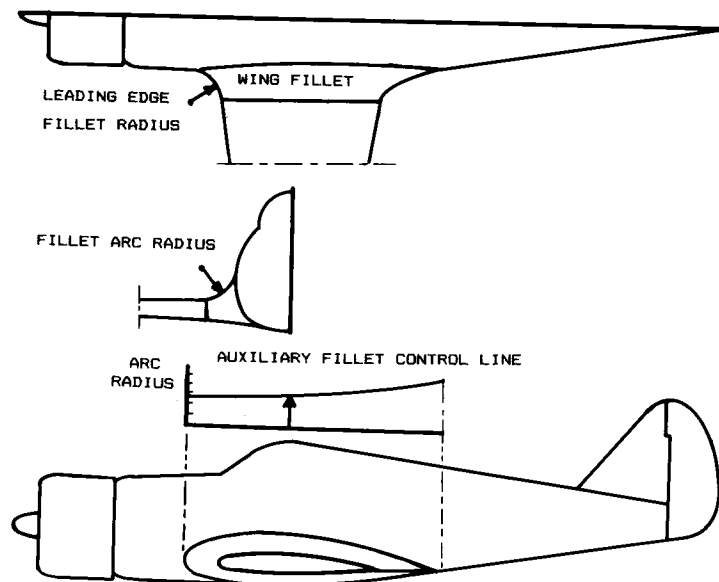


Fig. 7.32 Wing fillet layout.

of the wing. This avoids the necessity of constructing auxiliary views perpendicular to the wing surface to accurately develop each fillet radius.

7.9 AIRCRAFT LAYOUT PROCEDURES

There are really no standard procedures for aircraft layout. Every aircraft demands its own procedure. The designer must develop the aircraft drawing while simultaneously considering a wide variety of requirements, design drivers, and good design practices. This process can only be learned through practice and a high level of desire.

Nevertheless, certain generalities can be made. First, the initial design should be "roughed-out" before a quality drawing is begun. This is best done in the desired scale of the finished drawing. Common drawing scales are 1/10, 1/20, 1/40, and 1/100, depending upon the size of the aircraft. Computer-aided design systems usually work in "full scale" on the scope, and any desired scale can be selected if a paper copy is required.

This "roughed-out" design is based upon the conceptual sketch and should identify the locations of the major internal components such as the crew station, payload, passengers, fuel tanks, landing gear, and engines. The fuselage length should be based upon prior aircraft or the statistical approach presented earlier, and should be modified as necessary to contain the internal components.

The completed roughed-out layout is used to define the initial lofting approach. The designer must determine the number of longitudinal control lines to use as well as selecting the control stations. A minimum number of control stations should be selected to insure that all of the large internal components can be properly enclosed by the aircraft surface. Remember that the more control stations selected, the greater the difficulty in insuring smooth longitudinal contours.

At this point the actual layout is begun, using the roughed-out layout as an "underlay" (i.e., the actual layout is done on transparent paper or mylar placed over the roughed-out layout). The fuselage is developed using the selected lofting approach, and the internal components are drawn.

The wing and tail trapezoidal geometries should be drawn on separate pieces of paper. The mean aerodynamic chord should be shown, including the desired initial location of the wing with respect to the center of gravity.

The wing drawing can then be slid under the actual drawing and moved to the desired position with respect to the estimated center of gravity location on the fuselage. The trapezoidal wing geometry is traced on to the layout, and the actual wing geometry is drawn. This includes desired planform modifications as well as fillets and wing tips.

Tails are usually drawn after the fuselage is defined on the drawing. If the actual tail moment arm is different from that assumed in sizing the tail, the tail area should be recalculated at this time.

The wing and tail geometric parameters should be tabulated somewhere on the drawing, along with the estimated takeoff gross weight, fuel weight and volume, engine type and size (if not 100%), inlet capture area, propeller geometry, etc. This information will greatly aid those who later attempt to analyze the drawing.

7.10 WETTED AREA DETERMINATION

Aircraft wetted area (S_{wet}), the total exposed surface area, can be visualized as the area of the external parts of the aircraft that would get wet if it were dipped into water. The wetted area must be calculated for drag estimation, as it is the major contributor to friction drag.

The wing and tail wetted areas can be approximated from their planforms, as shown in Fig. 7.33. The wetted area is estimated by multiplying the true-view exposed planform area ($S_{exposed}$) times a factor based upon the wing or tail thickness ratio.

If a wing or tail were paper-thin, the wetted area would be exactly twice the true planform area (i.e., top and bottom). The effect of finite thickness is to increase the wetted area, as approximated by Eqs. (7.10) or (7.11). Note that the true exposed planform area is the projected (top-view) area divided by the cosine of the dihedral angle.

If $t/c < .05$

$$S_{wet} = 2.003 S_{exposed} \quad (7.10)$$

If $t/c > .05$

$$S_{wet} = S_{exposed} [1.977 + 0.52(t/c)] \quad (7.11)$$

The exposed area shown in Fig. 7.33 can be measured from the drawing in several ways. A professional designer will have access to a "planimeter," a mechanical device for measuring areas. Use of the planimeter is a dying art as the computer replaces the drafting board. Alternatively, the area can be measured by tracing onto graph paper and "counting squares."

The wetted area of the fuselage can be initially estimated using just the side and top views of the aircraft by the method shown in Fig. 7.34. The side- and top-view projected areas of the fuselage are measured from the drawing, and the values are averaged.

For a long, thin body circular in cross section, this average projected area times π will yield the surface wetted area. If the body is rectangular in cross section, the wetted area will be four times the average projected area. For typical aircraft, Eq. (7.12) provides a reasonable approximation.

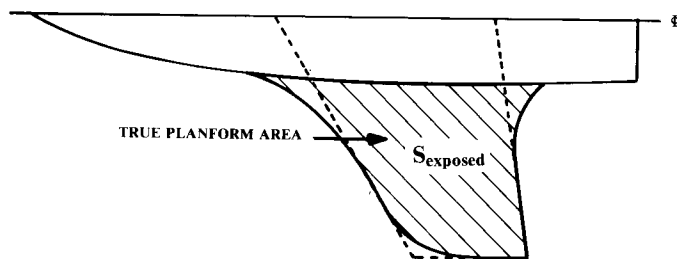


Fig. 7.33 Wing/tail wetted area estimate.

$$S_{wet} = K (A_{top} + A_{side})/2$$

$$K = \pi \text{ FOR ELLIPTIC CROSS SECTION}$$

$$K = 4 \text{ FOR SQUARE CROSS SECTION}$$

TYPICALLY USE $K = 3.4$

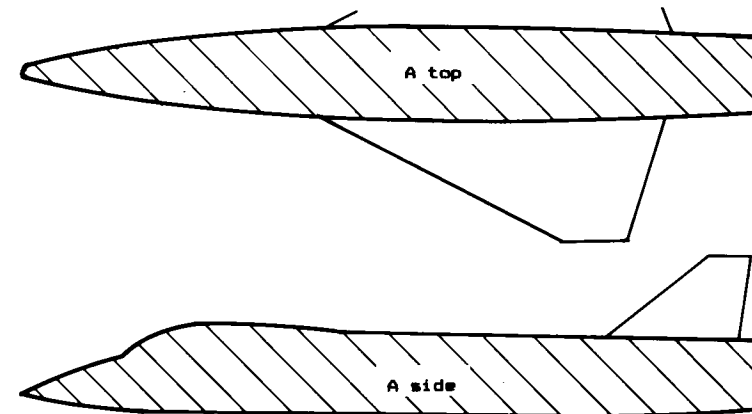


Fig. 7.34 Quick fuselage wetted area estimate.

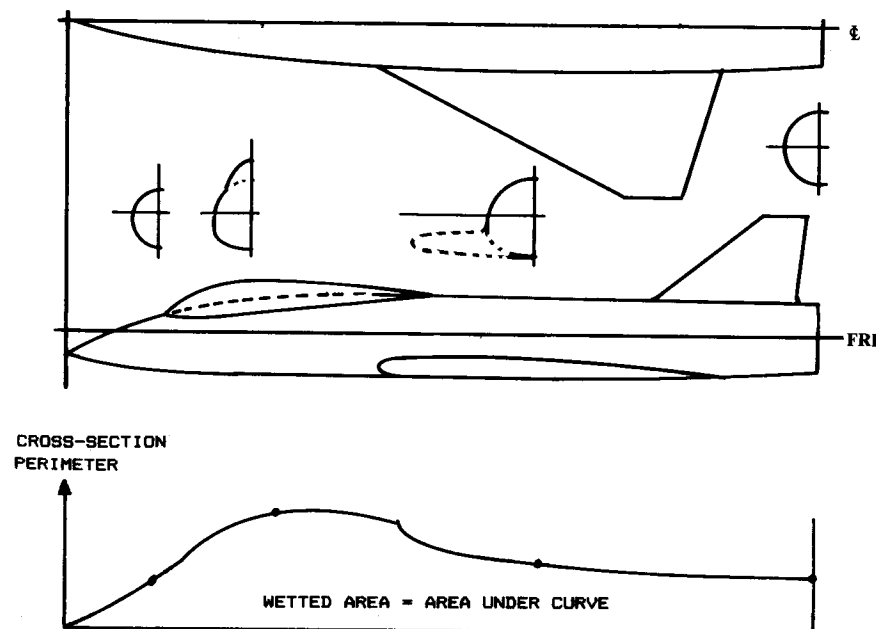


Fig. 7.35 Fuselage wetted area plot.

$$S_{\text{wet}} \cong 3.4 \left(\frac{A_{\text{top}} + A_{\text{side}}}{2} \right) \quad (7.12)$$

A more accurate estimation of wetted area can be obtained by graphical integration using a number of fuselage cross sections. If the perimeters of the cross sections are measured and plotted vs longitudinal location, using the same units on the graph, then the integrated area under the resulting curve gives the wetted area (Fig. 7.35).

Perimeters can be measured using a professional's "map-measure," or approximated using a piece of scrap paper. Simply follow around the perimeter of the cross section making tic marks on the paper, and then measure the total length using a ruler.

Note that the cross-sectional perimeter measurements should not include the portions where components join, such as at the wing-fuselage intersection. These areas are not "wetted."

7.11 VOLUME DETERMINATION

The aircraft internal volume can be used as a measure of the reasonableness of a new design, by comparing the volume to existing aircraft of similar weight and type. This is frequently done by customer engineering groups, using statistical data bases which correlate internal volume with takeoff gross weight for different classes of aircraft. An aircraft with a less-than-typical internal volume will probably be tightly packed, which makes for poor maintainability.

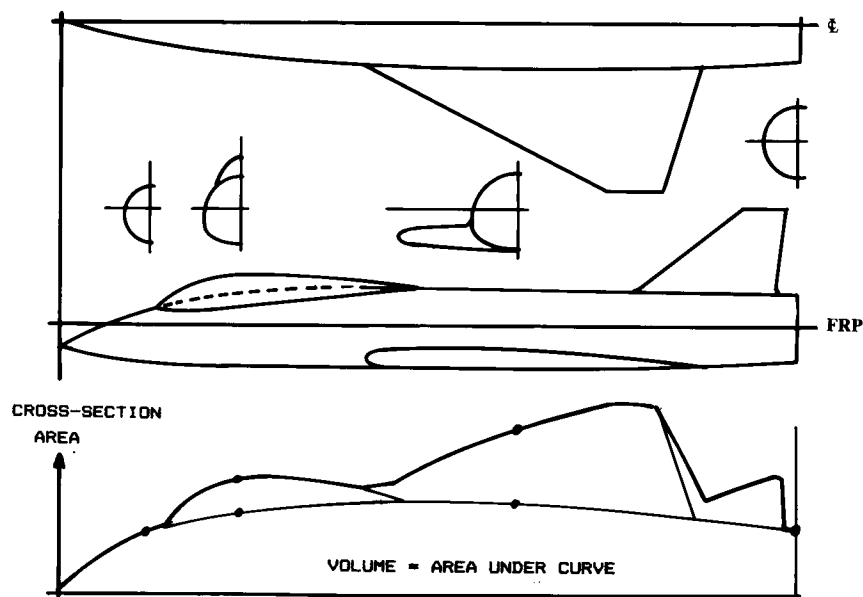


Fig. 7.36 Aircraft volume plot.

Aircraft internal volume can be estimated in a similar fashion to the wetted-area estimation. A crude estimate of the fuselage internal volume can be made using Eq. (7.13), which uses the side and top view projected areas as used in Eq. (7.12). "L" in Eq. (7.13) is the fuselage length.

$$\text{Vol} \cong 3.4 \frac{(A_{\text{top}})(A_{\text{side}})}{4L} \quad (7.13)$$

A more accurate estimate of internal volume can be found by a graphical integration process much like that used for wetted area determination. The cross-section areas of a number of cross sections are measured and plotted vs longitudinal location, using consistent units (typically inches horizontally and square inches vertically on the graph). The area under the resulting curve is the volume, as shown in Fig. 7.36.

To obtain reasonable accuracy, cross sections should be plotted and measured anywhere that the cross-sectional area changes substantially. This typically includes the start of an inlet duct, the start and end of a canopy, and where a wing or tail begins and ends.

Another use of the "volume distribution plot" is to predict and minimize supersonic wave drag and transonic drag rise. This will be discussed in Chapter 12.